

No.	Solution						
2(b)(ii)	<table><tr><th>$\Delta U/\text{J}$</th><th>q/J</th><th>w/J</th></tr><tr><td>0</td><td>-1.10×10^5</td><td>1.10×10^5</td></tr></table> Energy supplied = $IVt = 12 \times 230 \times 40 = 110400 \approx 110000 \text{ J}$	$\Delta U/\text{J}$	q/J	w/J	0	-1.10×10^5	1.10×10^5
$\Delta U/\text{J}$	q/J	w/J					
0	-1.10×10^5	1.10×10^5					
3(a)(i)	$\omega = 2\pi f = 2\pi\left(\frac{1200}{60}\right) = 2\pi(20)$ $= 125.66$ $= 126 \text{ rad s}^{-1}$						
3(a)(ii)	The difference between the normal contact force and its weight provides the						